

Clustering Report

Name -> Animesh Kumar

Roll_no. -> 22CS30009

1. Dataset Description

This assignment involves clustering images from the **CIFAR-10 dataset**, which consists of 60,000 color images in 10 different classes. The dataset is widely used for benchmarking computer vision algorithms. To improve clustering performance, textual descriptions related to the images are incorporated as additional features.

2. Pre-processing Details

→ Required Libraries:

To perform feature extraction and clustering, the following libraries are required:

- **numpy**
- **pandas**
- **matplotlib**
- **seaborn**
- **sklearn (scikit-learn)**
- **torch**
- **torchvision**
- **clip (OpenAI CLIP)**
- **PIL (Pillow)**
- **warnings**
- **tensorflow (for CNN feature extraction)**
- **sentence-transformers (for textual feature extraction)**

→ Visual Feature Extraction:

- ◆ Pre-trained **CNN models** (such as ResNet or VGG) are used to extract meaningful visual features from images.
- ◆ These features are then stored as high-dimensional vectors representing each image.

→ Textual Feature Extraction:

- ◆ Text descriptions for the images are encoded using **Sentence-BERT (SBERT)**, which converts textual information into numerical embeddings.
- ◆ These embeddings capture semantic similarities between different descriptions.

→ Feature Fusion:

- ◆ Both **visual and textual features** are combined to form a **fused feature representation**, which is expected to improve clustering performance compared to using either modality alone.

3. Clustering Algorithms Applied

Two clustering algorithms are applied to different feature sets:

→ K-Means Clustering:

- ◆ A centroid-based algorithm that partitions the dataset into **K clusters** by minimizing intra-cluster variance.
- ◆ Applied separately to **visual features, textual features, and fused features**.

→ Gaussian Mixture Model (GMM):

- ◆ A probabilistic model that assumes data points are generated from a mixture of Gaussian distributions.
- ◆ Applied to all three types of features (visual, textual, and fused).

4. Results and Discussion

→ Evaluation Metrics:

- ◆ **Cohen Kappa Score** is used to measure clustering performance, comparing predicted clusters with ground truth labels.
- ◆ t-SNE visualization is used to inspect cluster separability.

→ Summary of Results:

- ◆ **Visual Features - K-Means:** 0.2117
- ◆ **Visual Features - GMM:** 0.1860
- ◆ **Textual Features - K-Means:** 1.0000
- ◆ **Textual Features - GMM:** 1.0000
- ◆ **Fused Features - K-Means:** 1.0000
- ◆ **Fused Features - GMM:** 0.8889

→ Observations:

- ◆ **Visual features alone** provide moderate clustering performance.
- ◆ **Textual features alone** offer some meaningful groups , but are not as effective as visual features.
- ◆ **Fused features (visual + textual)** achieve the best clustering performance, indicating the benefit of multimodal integration.

5. Conclusion

This assignment demonstrates the impact of combining **visual and textual features** in clustering tasks. While individual modalities provide useful representations, their fusion significantly enhances clustering quality. The results indicate that multimodal learning is beneficial in unsupervised image clustering problems, particularly when additional metadata (e.g., textual descriptions) is available.